

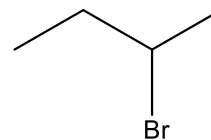
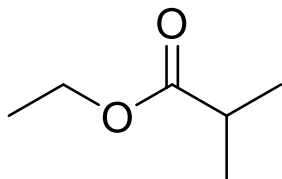
1. What is the energy difference between two spin states at the magnetic field 5.87 T of
- ^1H (given $\gamma = 267.512 \times 10^6 \text{ rad T}^{-1}\text{s}^{-1}$)
 - ^{13}C (given $\gamma = 67.2640 \times 10^6 \text{ rad T}^{-1}\text{s}^{-1}$)

~~2. At what magnetic field strength do proton (^1H) precesses at the frequency 300 MHz.
— Given that for ^1H , $\gamma = 267.512 \times 10^6 \text{ rad T}^{-1}\text{s}^{-1}$~~

3. a) If the observed chemical shift of a proton is 240 Hz from TMS, then find the chemical shift in ppm.
Given that instrument frequency is 60 MHz.
- b) The same sample is measured in 300 MHz instrument, then what will be chemical shift of the proton in ppm and Hz.
(CSIR NET)

4. The ^1H NMR spectrum of a dilute solution of a mixture of acetone and dichloromethane in CDCl_3 exhibits two singlet of 1:1 intensity. Molar ratio of acetone to dichloromethane in the solution is ? (CSIR NET)

5. Predict the no. of proton NMR peak will be observed in each of the molecule:



6. A compound with molecular formula C₃H₆O₂ has two peaks (both singlet) in the proton NMR spectra. Predict the structure.
¹H : 2.19 (3H), 3.66 (3H)

Q. Isomeric compounds **A** and **B** have the composition $C_{11}H_{12}O_4$. Deduce structures for **A** and **B** from the spectral data (in the proton NMR, ortho and meta couplings are visible). **Compound A**: IR (KBr): 1720 (s), 1240 (s) cm^{-1} , 1H NMR ($CDCl_3$): δ 2.46 (s, 3H), 3.94 (s, 6H), 8.05 (d, J 2 Hz, 2H), 8.49 (t, J 2 Hz, 1H); **Compound B**: IR (Nujol[®]): 1720 (s), 1245 (s) cm^{-1} , 1H NMR ($CDCl_3$): δ 2.63 (s, 3H), 3.91 (s, 6H), 7.28 (d, J 8 Hz, 1H), 8.00 (dd, J 8 Hz, 2 Hz; 1H), 8.52 (d, J 2 Hz, 1H). Assign all protons to the appropriate peaks.

2. At what magnetic field strength do proton (^1H) precesses at the frequency 300 MHz.
Given that for ^1H , $\gamma = 267.512 \times 10^6 \text{ rad T}^{-1} \text{ s}^{-1}$

Ans .

$$B_0 = \frac{2\pi\nu}{\gamma}$$

Given, $\gamma = 267.512 \times 10^6 \text{ rad T}^{-1} \text{ s}^{-1}$

$$\nu = 300 \times 10^6 \text{ s}^{-1}$$

$$\pi = 3.14 \text{ rad}$$

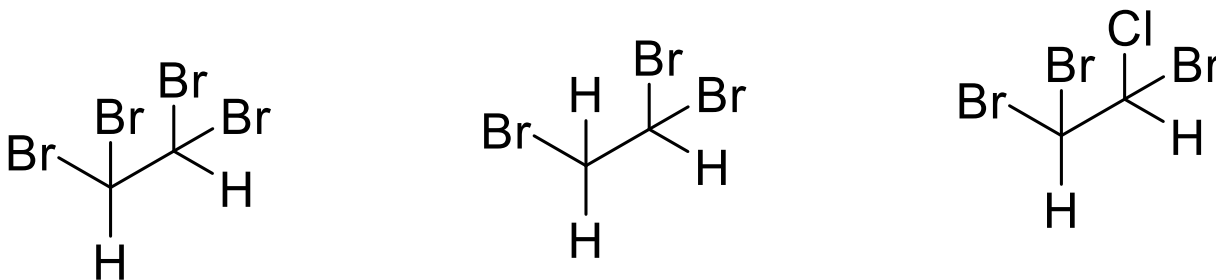
Putting the values in equation,

$$B_0 = \frac{2(3.14 \text{ rad}) \times 300 \times 10^6 \text{ s}^{-1}}{267.512 \times 10^6 \text{ rad T}^{-1} \text{ s}^{-1}}$$

$$B_0 = 7.04 \text{ T}$$

1. In the 400 MHz ^1H NMR spectrum of an organic compound, a doublet is observed. The two lines of the doublet appear at δ 2.35 and 2.38 ppm. Calculate the coupling constant (J) value.

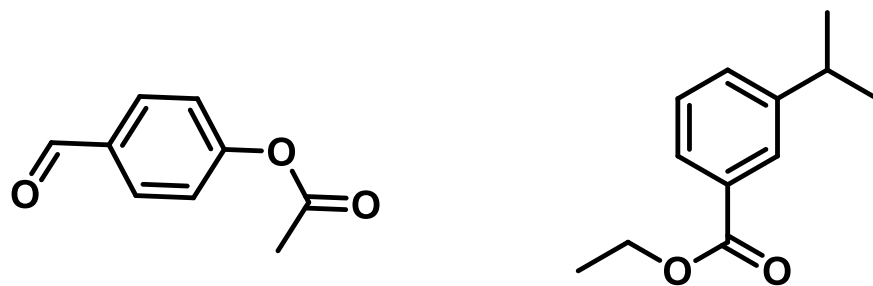
2. Write down the no. of different types of proton present in each molecule and then show their splitting.



3. ^1H NMR spectrum of an organic compound recorded on a 500 MHz spectrometer showed a quartet with line positions at 1759, 1753, 1747, 1741 Hz. Find the chemical shift (δ) and coupling constant (Hz) of the quartet.

(CSIR NET)

4. Draw the NMR of the compounds with approximate δ value in correct order with labelling and proper splitting.

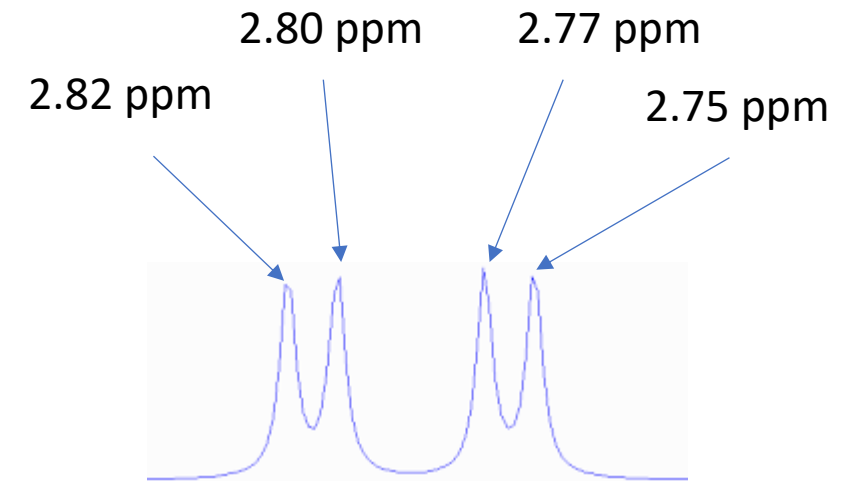


5. An organic compound ($C_{10}H_{12}O_2$) exhibits the following 1H NMR spectra data. Predict the structure

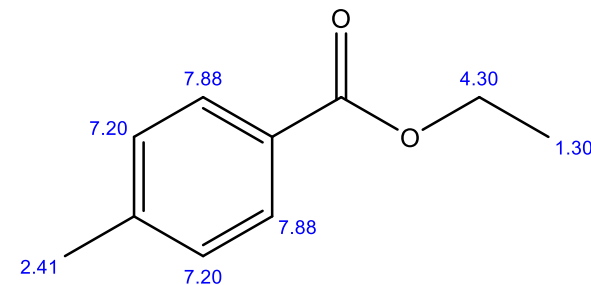
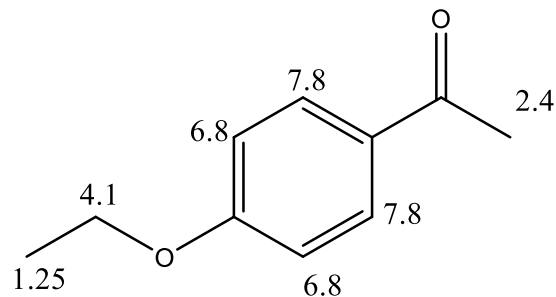
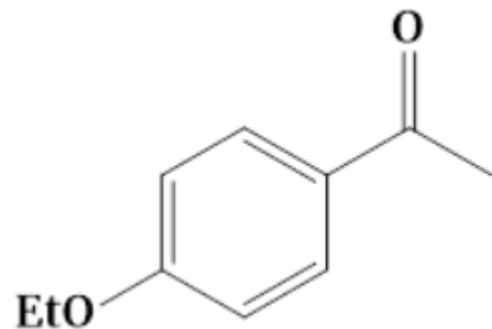
δ 7.80 (2H, d, $J = 8$ Hz), 6.80 (2H, d, $J = 8$ Hz), 4.10 (2H, q, $J = 7.2$ Hz), 2.4 (3H, s), 1.25 (3H, t, $J = 7.2$ Hz)

(GATE)

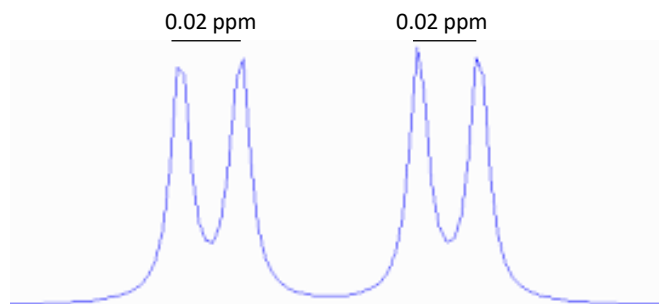
6. One proton peak split in doublet of doublet (as shown) due to presence of two types of neighbouring proton. Calculate the coupling constants.
Given that the instrument frequency is 400 MHz. The delta value of each peak
Shown by arrow.



5.



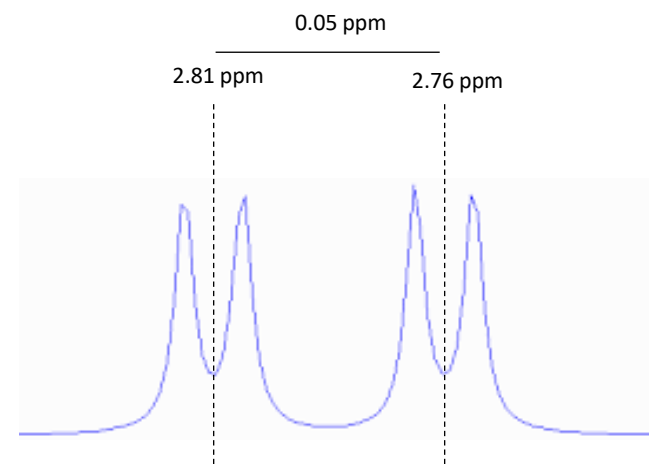
6.



$$2.82 - 2.80 = 0.02 \text{ ppm}$$

$$2.77 - 2.75 = 0.02 \text{ ppm}$$

$$\text{Coupling constant (J)} = \frac{0.02 \times 400 \times 10^6}{10^6} = 8 \text{ Hz}$$

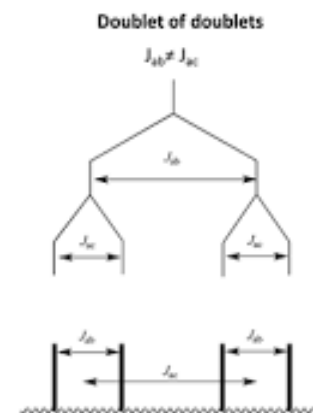


$$(2.82 + 2.80)/2 = 2.81 \text{ ppm}$$

$$(2.77 + 2.75)/2 = 2.76 \text{ ppm}$$

$$2.81 - 2.76 = 0.05 \text{ ppm}$$

$$\text{Coupling constant (J)} = \frac{0.05 \times 400 \times 10^6}{10^6} = 20 \text{ Hz}$$



1. An organic compound ($C_7H_{12}O_2$) exhibited the following data in the 1H NMR spectrum.

δ 7.10 (1H, dt, $J = 16$ Hz and 7.2 Hz), 5.90 (1H, dt, $J = 16$ Hz and 2 Hz), 4.1 (2H, q, $J = 7.2$ Hz), 2.10 (2H, m), 1.25 (3H, t, $J = 7.2$ Hz), 0.90 (3H, t, $J = 7.2$ Hz) ppm. (CSIR NET)

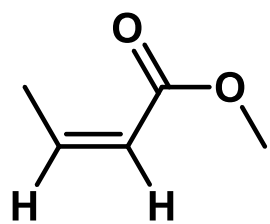
Q2. Determine the correct structure of a compound having molecular formula $C_6H_{11}BrO_2$, based on the following characteristic spectral data:

IR: 1736 cm^{-1}

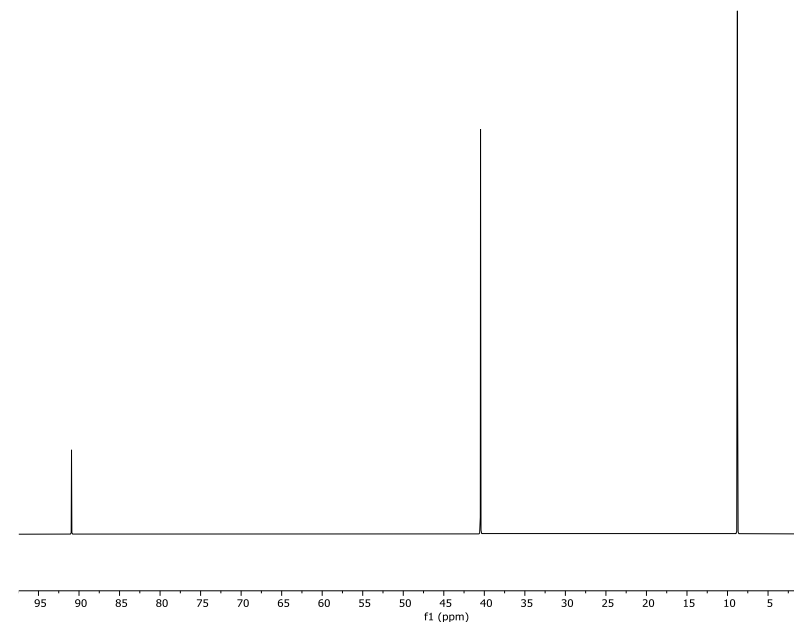
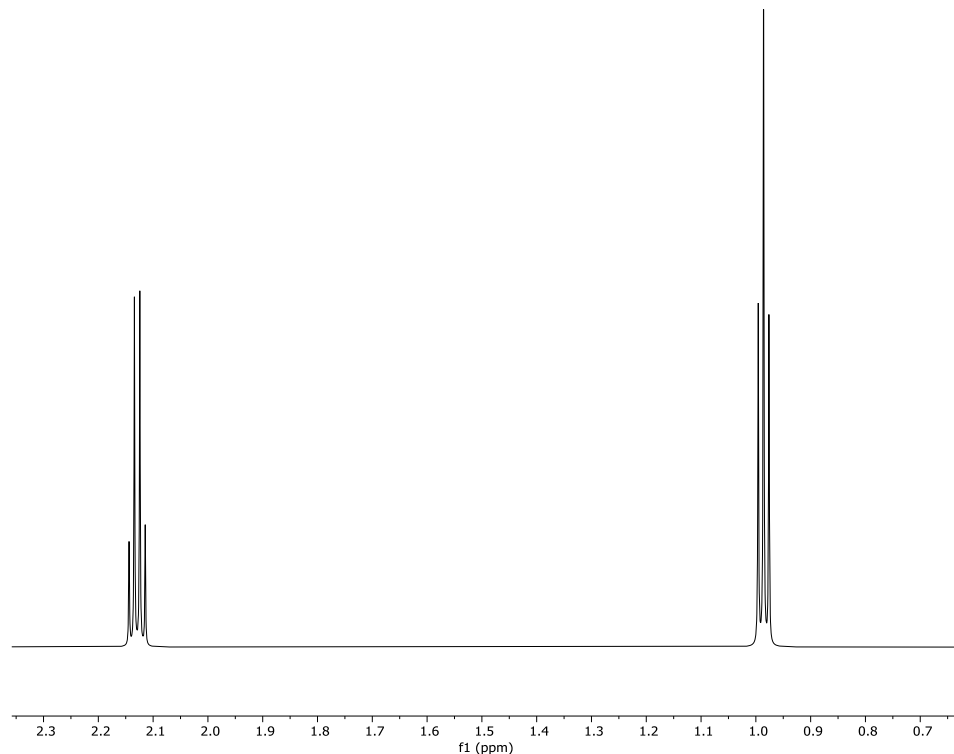
1H NMR (δ , ppm): 3.59 (s, 3H), 3.32 (t, 2H), 2.25 (t, 2H), 1.85–1.75 (m, 2H), 1.73–1.62 (m, 2H)

^{13}C NMR (δ , ppm): 174.0, 51.0, 32.9, 32.8, 31.0, 23.0 (CSIR NET)

Q3. Predict the number of signals expected in the 1H NMR spectrum of the following compound and assign order of chemical shift (δ) values to each type of proton. (CSIR NET)



4. The ^1H and ^{13}C NMR data is provided of one compound having formula $(\text{C}_5\text{H}_{10}\text{Cl}_2)$. Predict the structure of the compound.



5. Two isomers with the formula $\text{C}_{10}\text{H}_{12}\text{O}$. Predict the structure of the compounds.

Compound A: IR (neat): 1675 cm^{-1} ; ^1H NMR (CDCl_3): 1.20 (d, $J = 7\text{ Hz}$, 6H), 3.53 (septet, $J = 7\text{ Hz}$, 1H), 7.94 (m, 2H), 7.68 (m, 1H), 7.3 (m, 2H).

Compound B: IR (neat): 1705 cm^{-1} ; ^1H NMR (CDCl_3): 0.95 (t, $J = 7\text{ Hz}$, 3H), 2.35 (q, $J = 7\text{ Hz}$, 2H), 3.6 (s, 2H), 7.20 (m, 5H).

Q 6. Match the following with ^{13}C NMR value:

Compound

^{13}C NMR chemical shift (δ ppm)

(A) Acetic acid

(i) 95

(B) Acetonitrile

(ii) 115

(C) Acetone

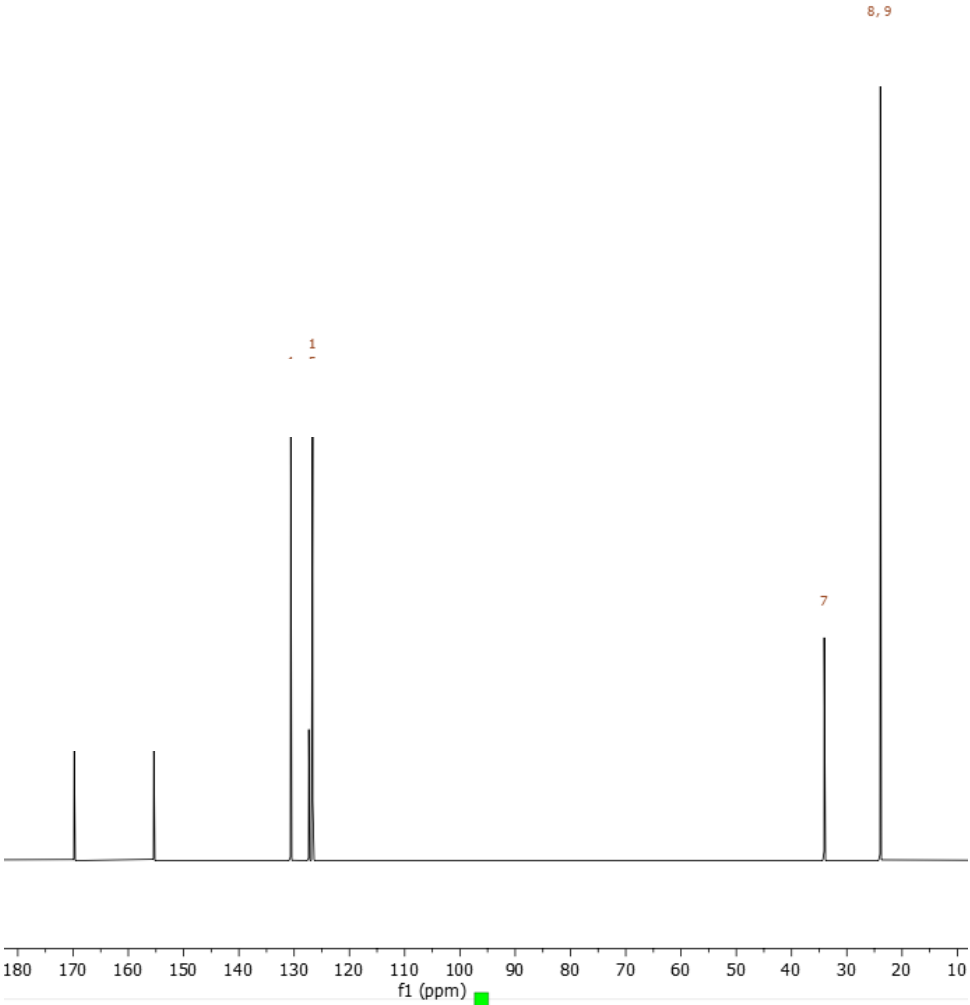
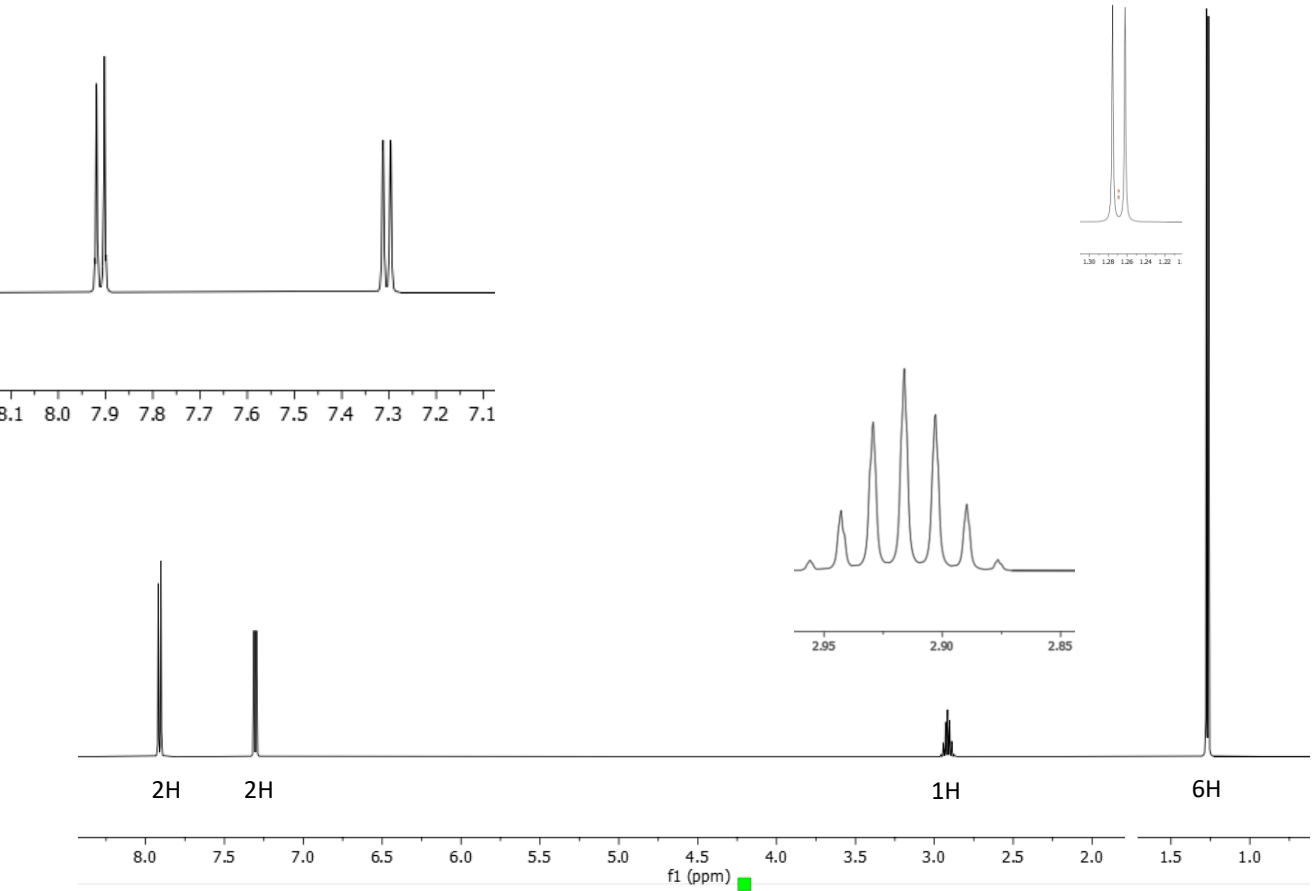
(iii) 175

(D) Carbon tetrachloride

(iv) 205

(CSIR NET)

5. ^1H and ^{13}C NMR is given of a compound with molecular formula $\text{C}_{10}\text{H}_{12}\text{O}_2$ and it has one exchangeable proton near 12 ppm . Justify the DEPT results. What is the structure of the compound.



Normal Carbon	DEPT-135	DEPT-90
26	Positive	No peak
34	Positive	Positive
126.6	Positive	Positive
130.5	Positive	Positive

Solution in next page.

5. ^1H and ^{13}C NMR is given of a compound with molecular formula $\text{C}_{10}\text{H}_{12}\text{O}_2$ and it has one exchangeable proton near 12 ppm . Justify the DEPT results. What is the structure of the compound.

